



GONZAGA, SOFIA LORRAINE
BSIT 4A

ITEP 414: System Administration and Maintenance
Week 3: Enterprise Server Deployment and Operating System Installation

Part C – BIOS vs UEFI Comparison

Comparison Point	Legacy BIOS	UEFI (Unified Extensible Firmware Interface)
Definition	Basic Input/Output System; legacy firmware initializing hardware on startup.	Modern firmware architecture designed as a faster, secure replacement for BIOS.
Boot Process	Loads 16-bit real-mode code directly from the Master Boot Record (MBR) sector.	Directly executes 32-bit or 64-bit EFI executable files (.efi) from the EFI System Partition.
Maximum Disk Support	Supports disks up to 2 TB.	Supports disks up to 9.4 ZB (Zettabytes).
Partition Style	MBR (Master Boot Record) — max 4 primary partitions.	GPT (GUID Partition Table) — up to 128 primary partitions natively.
Security Features	Basic password protection; vulnerable to bootkits/rootkits.	Secure Boot, cryptographic signature verification for drivers/bootloaders.
Speed	Slower due to sequential hardware testing and 16-bit execution constraints.	Faster due to parallel hardware initialization and native 32/64-bit operation.
Advantages	Universal support on legacy hardware, simple architecture.	Handles huge drives, high security, fast boot times, modular driver support.
Disadvantages	Limited drive sizes, slow boot times, vulnerable to low-level malware.	More complex firmware setup; requires GPT partition scheme.
Modern Usage	Deprecated; rarely used on systems manufactured after 2020.	Standard industry default for modern servers, PCs, and cloud virtualization platforms.

Why UEFI Replaced Legacy BIOS in Modern Computers

Legacy BIOS was designed in the late 1970s for IBM PCs operating in 16-bit real mode with limited memory addressing. As server and workstation hardware advanced, the technical architecture of BIOS created significant system bottlenecks:

1. **Storage Capacity Limits:** BIOS relies on the MBR partition scheme, which uses 32-bit sector addressing. This caps usable storage drives at 2 TB. Modern enterprise servers routinely deploy storage arrays exceeding dozens or hundreds of terabytes, requiring the GUID Partition Table (GPT) natively supported by UEFI.
2. **Pre-Boot Security (Secure Boot):** BIOS lacks authentication mechanisms before the operating system kernel loads, making it vulnerable to persistent rootkits and bootkit exploits. UEFI introduces **Secure Boot**, verifying cryptographic signatures of bootloaders (like GRUB) and kernel drivers before execution to ensure system integrity.
3. **Performance and Initialization Speed:** BIOS tests and initializes hardware components sequentially in 16-bit mode. UEFI operates natively in 32-bit or 64-bit mode, enabling parallel device initialization, modular EFI driver loading, and drastically reduced system boot times.

Conclusively, UEFI provides the speed, storage flexibility, and low-level security required for modern enterprise environments, making BIOS obsolete for current deployment standards.